

## SAFETY DATA SHEET

Creation Date 30-Nov-2010

Revision Date 29-Mar-2024

Revision Number 5

### 1. Identification

**Product Name** Phosphorus lump, red

**Cat No. :** 00248

**CAS No** 7723-14-0  
**Synonyms** None

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable solids

Category 1

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable solid

**Precautionary Statements****Prevention**

Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Wear protective gloves/protective clothing/eye protection/face protection

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Hazards not otherwise classified (HNOC)**

Harmful to aquatic life with long lasting effects  
Explosive when mixed with oxidizing substances

### 3. Composition/Information on Ingredients

| Component  | CAS No    | Weight % |
|------------|-----------|----------|
| Phosphorus | 7723-14-0 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                            |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                                                                      |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air.                                                                                                                                       |
| <b>Ingestion</b>                           | Never give anything by mouth to an unconscious person. Drink plenty of water. Induce vomiting, but only if victim is fully conscious. Call a physician immediately. Get medical attention. |
| <b>Most important symptoms and effects</b> | No information available.                                                                                                                                                                  |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                      |

### 5. Fire-fighting measures

|                                       |                                       |
|---------------------------------------|---------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Dry chemical, soda ash, lime or sand. |
| <b>Unsuitable Extinguishing Media</b> | Carbon dioxide (CO <sub>2</sub> )     |
| <b>Flash Point</b>                    | No information available              |
| <b>Method -</b>                       | No information available              |
| <b>Autoignition Temperature</b>       | 215 °C                                |
| <b>Explosion Limits</b>               |                                       |
| <b>Upper</b>                          | No data available                     |
| <b>Lower</b>                          | No data available                     |

**Sensitivity to Mechanical Impact** No information available  
**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Combustible material.

**Hazardous Combustion Products**

Oxides of phosphorus. Phosphorus trihydride (phosphine).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
0

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Avoid dust formation. Use spark-proof tools and explosion-proof equipment. Provide adequate ventilation. Sweep up and shovel into suitable containers for disposal.

**7. Handling and storage****Handling**

Avoid contact with skin and eyes. Avoid contact with skin and clothing. Avoid breathing vapors or mists. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep away from oxidizing agents. Incompatible Materials. Halogens. Metals. copper. Oxidizing agent.

**8. Exposure controls / personal protection****Exposure Guidelines**

| Component  | ACGIH TLV | OSHA PEL                             | NIOSH | Mexico OEL (TWA) |
|------------|-----------|--------------------------------------|-------|------------------|
| Phosphorus |           | (Vacated) TWA: 0.1 mg/m <sup>3</sup> |       |                  |

**Legend**

OSHA - Occupational Safety and Health Administration

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eyeface Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.

**Recommended Filter type:**

Particulates filter conforming to EN 143.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Solid                    |
| Appearance                             | Dark red                 |
| Odor                                   | Slight                   |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | 590 °C / 1094 °F         |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 215 °C                   |
| Decomposition Temperature              | 300°C                    |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | P                        |
| Molecular Weight                       | 30.96                    |

## 10. Stability and reactivity

|                                  |                                                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | No                                                                                                                       |
| Stability                        | Stable under normal conditions.                                                                                          |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Excess heat. Exposure to light. Incompatible products. |
| Incompatible Materials           | Halogens, Metals, copper, Oxidizing agent                                                                                |
| Hazardous Decomposition Products | Oxides of phosphorus, Phosphorus trihydride (phosphine)                                                                  |
| Hazardous Polymerization         | No information available.                                                                                                |
| Hazardous Reactions              | None under normal processing.                                                                                            |

## 11. Toxicological information

## Acute Toxicity

## Product Information

## Component Information

| Component  | LD50 Oral                 | LD50 Dermal | LC50 Inhalation             |
|------------|---------------------------|-------------|-----------------------------|
| Phosphorus | >15000 mg/kg (Rat Female) | Not listed  | LC50 = 4.3 mg/L ( Rat ) 1 h |

Toxicologically Synergistic Products No information available

## Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component  | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------|-----------|------------|------------|------------|------------|------------|
| Phosphorus | 7723-14-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component  | Freshwater Algae | Freshwater Fish                   | Microtox   | Water Flea          |
|------------|------------------|-----------------------------------|------------|---------------------|
| Phosphorus | Not listed       | LC50: 33.2 mg/L/96h (Danio rerio) | Not listed | EC50: 10.5 mg/L/48h |

**Persistence and Degradability** Insoluble in water

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1338  
 Proper Shipping Name PHOSPHORUS, AMORPHOUS  
 Hazard Class 4.1  
 Packing Group III

### TDG

UN-No UN1338  
 Proper Shipping Name PHOSPHORUS, AMORPHOUS  
 Hazard Class 4.1  
 Packing Group III

### IATA

UN-No UN1338  
 Proper Shipping Name PHOSPHORUS, AMORPHOUS

|                             |                       |
|-----------------------------|-----------------------|
| <b>Hazard Class</b>         | 4.1                   |
| <b>Packing Group</b>        | III                   |
| <b>IMDG/IMO</b>             |                       |
| <b>UN-No</b>                | UN1338                |
| <b>Proper Shipping Name</b> | PHOSPHORUS, AMORPHOUS |
| <b>Hazard Class</b>         | 4.1                   |
| <b>Packing Group</b>        | III                   |

## 15. Regulatory information

### United States of America Inventory

| Component  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------|-----------|------|-----------------------------------------------|-----------------------------|
| Phosphorus | 7723-14-0 | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

### TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)

Not applicable

### TSCA 12(b) - Notices of Export

Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component  | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Phosphorus | 7723-14-0 | X   | -    | 231-768-7 | X     | X    |      | X    | X     | KE-28713 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component  | CAS No    | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|------------|-----------|----------|-------------------------------|---------------------------------|
| Phosphorus | 7723-14-0 | >95      | 1.0 %                         | -                               |

### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

### CWA (Clean Water Act)

| Component  | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Phosphorus | X                          | 1 lb                        | -                      | -                         |

### Clean Air Act

| Component  | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|------------|-----------|-------------------------|-------------------------|
| Phosphorus | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component  | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|------------|--------------------------|-------------------------------------------|-------------------------------|
| Phosphorus | 1 lb                     | 1 lb                                      | 1 lb<br>0.454 kg              |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component  | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------|---------------|------------|--------------|----------|--------------|
| Phosphorus | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product contains the following DHS chemicals:

**Legend** - STQs = Screening Threshold Quantities, APA = A placarded amount

| Component  | DHS Chemical Facility Anti-Terrorism Standard |
|------------|-----------------------------------------------|
| Phosphorus | Theft STQs - 400lb                            |

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

| Component  | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phosphorus | 7723-14-0 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component  | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Phosphorus | 7723-14-0 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

## Other International Regulations

| Component  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Phosphorus | 7723-14-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 30-Nov-2010  
**Revision Date** 29-Mar-2024  
**Print Date** 29-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**